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Paper sheet transport device and paper sheet handling device

Abstract

The present invention enables determination as to whether a banknote **100** is to be discharged even when only the feature quantity of a portion of the banknote is acquired at the time of discharging the banknote in a case in which a transport route length L**1** from a reading position **18***a* of an optical recognition sensor **18** to an inlet/outlet **12** as a banknote discharge port is shorter than a total length LA of the banknote in a transport direction. At the time of discharging the banknote **100** already authenticated, a control unit **200** judges the type (denomination) of the banknote on the basis of a feature quantity of a reading target **103** that passes the optical recognition sensor **18** before a leading end **100***a* of the banknote **100** transported in a discharge direction protrudes to outside the device from the inlet/outlet **12.**

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Background/Summary

RELATED APPLICATIONS

(1) This application is the U.S. National Phase of and claims priority to International Patent Application No. PCT/JP2022/022937, International Filing Date Jun. 7, 2022, entitled Paper Sheet Transport Device And Paper Sheet Handling Device, which claims benefit of Japanese Application No. JP2021-128137 filed Aug. 4, 2021, both of which are incorporated herein by reference in their entireties.

FIELD

(2) The present invention relates to a paper sheet transport device and a paper sheet handing device that transport paper sheets such as banknotes.

BACKGROUND

(3) Patent Literature 1 describes a paper sheet handling device including a judgement/authentication unit that judges the denomination of a banknote, authenticity thereof, and whether the banknote is fit or unfit (whether the banknote is a fit note that is recirculatable or an unfit note that is not to be recirculated). The judgement/authentication unit includes a thickness detector and an optical detector along a transport route. The judgement/authentication unit performs judgement of the denomination of a banknote, authenticity thereof, and whether the banknote is fit or unfit on the basis of an output signal that is obtained at the time of passage of the banknote through each of the detectors. The banknote handling device changes the transport destination of a banknote having passed through the judgement/authentication unit according to a result of the judgement.

(4) In Patent Literature 1, the transport destination of a banknote is changed after the judgement of the banknote is performed based on the output signal obtained from the entire banknote.

CITATION LIST

Patent Literature

(5) Patent Literature 1: Japanese Patent Application Laid-open No. H1-209595

SUMMARY

Technical Problem

(6) When a banknote is transported in the discharge direction in a device in which the transport route length from a detector to a banknote discharge port is shorter than the total length of the banknote in the transport direction, the leading end of the banknote protrudes from the banknote discharge port before the detector reads information of the entire banknote. When the leading end of a banknote protrudes from the banknote discharge port, the banknote is likely to be pulled out. Accordingly, it is undesirable to cancel discharge of the banknote after the leading end of the banknote protrudes from the banknote discharge port.

(7) The present invention has been made in view of the above circumstances, and has an object to enable determination as to whether a paper sheet is to be discharged even when only the feature quantity of a portion of the paper sheet is acquired at the time of discharge of the paper sheet in a case in which the transport route length from a detector to the paper sheet discharge port is shorter than the total length of the paper sheet in the transport direction.

Solution to Problem

(8) In order to achieve the above object, a paper sheet transport device according to the present invention comprises a paper sheet discharge port for discharging a paper sheet to outside the device, a recognition sensor that is arranged on a transport route for the paper sheet toward the paper sheet discharge port to read a feature quantity of the paper sheet transported on the transport route, and a paper sheet type judging unit that judges a type of the paper sheet on a basis of the read feature quantity of the paper sheet, wherein a transport route length from the paper sheet discharge

port to the recognition sensor is configured to be shorter than a total length of the paper sheet in a transport direction, and the paper sheet type judging unit judges, at a time of discharging the paper sheet already authenticated, a type of the paper sheet on a basis of a feature quantity of a paper sheet portion passing the recognition sensor before a leading end of the paper sheet transported in a discharge direction protrudes to outside the device from the paper sheet discharge port.
Advantageous Effects of Invention
(9) At the time of discharging a paper sheet, whether to discharge the paper sheet can be determined even when only the feature quantity of a portion of the paper sheet is acquired.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a vertical sectional view illustrating an internal configuration of an entire banknote transport device according to one embodiment of the present invention.
(2) FIG. **2** is a block diagram of the banknote transport device illustrating a configuration related to control executed by a control unit.
(3) FIGS. **3**(*a*) to **3**(*c*) are schematic diagrams for explaining a relation between the internal configuration of the banknote transport device and a banknote being discharged.
(4) FIGS. **4**(*a*) and **4**(*b*) are schematic diagrams for explaining a relation between the internal configuration of the banknote transport device and a banknote being discharged.
(5) FIG. **5** is a flowchart illustrating an operation at the time of banknote deposit.
(6) FIG. **6** is a flowchart illustrating an operation at the time of banknote discharge.
DESCRIPTION OF EMBODIMENTS
(7) The present invention will be described below in detail with embodiments illustrated in the drawings. Constituent elements, types, combinations, shapes, and relative arrangements described in the following embodiments are merely explanatory examples, and are not intended to limit the scope of the present invention solely thereto unless otherwise specified.
First Embodiment
(8) FIG. **1** is a vertical sectional view illustrating an internal configuration of the entire banknote transport device according to one embodiment of the present invention.
(9) A banknote transport device (a paper sheet transport device) **1** according to one embodiment of the present invention includes an inlet/output (a paper sheet discharge port) **12** that discharges a banknote (a paper sheet) to outside the banknote transport device **1**, an optical recognition sensor (a recognition sensor) **18** that is arranged on a transport route **10** for a banknote toward the inlet/outlet **12** to read a feature quantity of a banknote transported on the transport route **10**, and a control unit (a paper sheet type judging unit) **200** that judges the type of a banknote on the basis of the read feature quantity of the banknote.
(10) As illustrated in FIG. **3**(*b*), a transport route length L**1** from the inlet/outlet **12** to the optical recognition sensor **18** in the banknote transport device **1** is configured to be shorter than a total length LA in the transport direction of a banknote **100** transported on the transport route **10**.
(11) In a dispense operation for the banknote **100**, the control unit **200** judges the type (denomination) of a banknote on the basis of the feature quantity acquired from a banknote portion (a reading target **103**) from a leading end **100***a* of the banknote **100** transported in the discharge direction (an arrow A**1** direction in FIG. **1**) to an intermediate part in the transport direction of the banknote **100**. Specifically, the control unit **200** judges the type of the banknote **100** on the basis of the feature quantity of the banknote portion (the reading target **103**) passing the optical recognition sensor **18** before the leading end **100***a* of the banknote **100** transported in the discharge direction protrudes outside the banknote transport device **1** from the inlet/outlet **12**.
(12) <Banknote Transport Device>

(13) An internal configuration of the banknote transport device is explained with reference to FIG. **1**. While a banknote is described as an example of a paper sheet in this example, the present device is also applicable to prevention of a fraudulent act in transport of a paper sheet other than a banknote, such as a security, a cash voucher, and a ticket.
(14) The banknote transport device **1** is used in a state of being attached to the body of a banknote handling device (a paper sheet handling device body) such as a banknote deposit machine, various vending machines, and a money changer (all not illustrated), and a banknote received in the banknote transport device **1** is subjected to recognition of authenticity and denomination of the banknote by a recognition sensor, and is then sequentially stored one by one in a cashbox (a banknote storage) in the banknote handling device body. The banknote handling device (the paper sheet handing device) is configured to include the banknote handling device body and the banknote transport device **1**.
(15) The banknote transport device **1** includes a lower unit **3**, and an upper unit **4** supported on the lower unit **3** to be capable of opening and closing, and the banknote transport route (transport route) **10** is formed between opposing surfaces of these units when the units are in a closed state illustrated in FIG. **1**.
(16) The inlet/outlet **12** for discharging the banknote **100** to the outside of the banknote transport device **1** or introducing the banknote **100** to the inside of the banknote transport device **1** is provided at one end of the transport route **10**. Inside the inlet/outlet **12**, an inlet/outlet tracking sensor **14** for detecting a banknote, an inlet/outlet roller pair **16**, the optical recognition sensor **18** that reads information for recognizing the denomination and the authenticity of a banknote, a relay roller pair **20**, a tracking sensor **22** on an one end side of a fraud prevention mechanism, a fraud prevention mechanism **24** including an opening/closing member **50** for fraud detection, a fraud preventing motor **120**, and the like, a tracking sensor **26** on the other end side of the fraud prevention mechanism, a communication port roller pair **28**, a communication port tracking sensor **30**, and a communication port **32** are arranged along the transport route **10**. A transport motor **35** that drives the roller pairs **16**, **20**, and **28** for transporting a banknote, and the control unit (a CPU, an MPU, a ROM, and a RAM) **200** that judges the denomination and the authenticity of a banknote on the basis of the recognition information from the optical recognition sensor **18** and that controls control targets such as the transport motor **35** on the basis of a banknote detection signal from each of the tracking sensors are also arranged.
(17) The control unit **200** is constituted of a unit including a CPU (Central Processing Unit), a ROM (Read Only Memory), and a RAM (Random Access Memory), or is constituted of an MPU (Micro Processing Unit) including necessary modules as well as the CPU, the ROM, and the RAM assembled into one chip, or the like. A banknote discharged from the communication port **32** is stored in a banknote storage (a paper sheet storage or a stacker device, not illustrated). The above configuration of the banknote transport device **1** is merely an example and various modifications can be made. For example, the number of used motors, arrangement of the roller pairs, the type of the recognition sensor, and the like can be variously changed or selected.
(18) Each of the roller pairs **16**, **20**, and **28** is constituted of a driving roller placed on the side of the lower unit **3** and a driven roller placed on the side of the upper unit **4**, and includes a configuration of transporting a banknote while nipping both surfaces of the banknote. The optical recognition sensor **18** is, for example, a photocoupler that is constituted of a light-emitting element and a light-receiving element arranged to oppose across the transport route **10**. The optical recognition sensor **18** having the above configuration can recognize an optical pattern (optical features) of a banknote by transmitting infrared light generated by the light-emitting element through the banknote and receiving the transmitted light. A magnetic sensor may be used as the recognition sensor. Alternatively, an image sensor (for example, a CIS; Contact Image Sensor) that recognizes an optical pattern of a banknote by receiving reflection light emitted from a light source and reflected by the banknote can be used as the recognition sensor.

(19) <Functional Configuration>
(20) FIG. **2** is a block diagram of the banknote transport device illustrating a configuration related to control executed by the control unit.
(21) The inlet/outlet tracking sensor **14**, the optical recognition sensor **18**, and the communication port tracking sensor **30** are connected to input terminals of the control unit **200**. The transport motor **35** is connected to an output terminal of the control unit **200**.
(22) The control unit **200** includes an authentication unit that receives an output of the optical recognition sensor **18** to judge whether a banknote is genuine, a banknote type judging unit (paper sheet type judging unit) that receives the output of the optical recognition sensor **18** to judge the type (denomination) of the banknote, and a driving control unit that controls driving of the transport motor **35**. The authentication unit, the banknote type judging unit, and the driving control unit are realized by the CPU included in the control unit, which reads a program from the ROM to load the program into the RAM, and executes the program.
(23) With reference to FIG. **1**, when the banknote **100** received from the inlet/outlet **12** in an arrow A**2** direction in FIG. **1** is judged as a genuine banknote, the control unit **200** executes control to continuously drive the transport motor **35** in a banknote receiving direction until the communication port tracking sensor **30** detects passage of a banknote, to transport the banknote to the banknote storage. When the banknote is not judged as a genuine banknote, the control unit **200** executes control to reversely rotate the transport motor **35** and continuously drive the transport motor **35** in a banknote discharge direction until the inlet/outlet tracking sensor **14** detects passage of the banknote, to return the banknote to the inlet/outlet **12**.
(24) When the banknote **100** received from the banknote storage through the communication port **32** is judged as a type of a banknote to be discharged from the inlet/outlet **12**, the control unit **200** executes control to continuously drive the transport motor **35** in the banknote discharge direction until the inlet/outlet tracking sensor **14** detects passage of the banknote, to discharge the banknote from the inlet/output **12**. When the banknote is not judged as a type of a banknote to be discharged from the inlet/outlet **12**, the control unit **200** executes control to reversely rotate the transport motor **35** and continuously drive the transport motor **35** in the banknote receiving direction until the communication port tracking sensor **30** detects passage of the banknote, to transport the banknote to the banknote storage.
(25) <Processing of Banknote being Discharged>
(26) FIGS. **3**(*a*) to **3**(*c*) and FIGS. **4**(*a*) and **4**(*b*) are schematic diagrams for explaining a relation between the internal configuration of the banknote transport device and a banknote being discharged. In the present example, the banknote **100** is transported in a direction along the long edge direction.
(27) As illustrated in the drawings, in the banknote transport device **1**, the transport route length L**1** from the optical recognition sensor **18** to the inlet/outlet **12** is shorter than the total length LA of the banknote **100** in the transport direction. More precisely, the transport route length L**1** is the length of the transport route **10** from a reading position **18***a* where the optical recognition sensor **18** reads information of a banknote to the inlet/outlet **12**.
(28) As illustrated in FIG. **3**(*a*), at the time of discharging the banknote **100**, a leading end portion **101** of the banknote **100** protrudes to the outside of the banknote transport device **1** from the inlet/outlet **12** before the optical recognition sensor **18** reads information of the total length LA of the banknote **100** in the transport direction. When the leading end portion **101** of the banknote **100** is protruded to the outside of the banknote transport device **1** from the inlet/outlet **12**, the banknote can be easily pulled out only by pinching the leading end portion **101** with fingers or the like. Therefore, even when the banknote **100** is a type of a banknote that is not a discharge target, it is undesirable to cancel discharge processing of the banknote **100** after the leading end portion **101** of the banknote **100** is protruded to the outside of the banknote transport device **1** from the inlet/outlet **12**.

(29) Therefore, in the present embodiment, a boundary position where easiness of pulling out a banknote transported in the discharge direction greatly changes is set as a discharge cancellation line B. When the banknote **100** is not a discharge target, the control unit **200** executes control to cancel the discharge processing of the banknote **100** before the leading end **100***a* of the banknote **100** reaches the discharge cancellation line B.
(30) FIG. **3**(*b*) is an example where the inlet/outlet **12** is the discharge cancellation line B. In this example, a transport direction length LB of a portion (the reading target **103**) of the banknote **100** that can be read by the optical recognition sensor **18** before the banknote **100** reaches the discharge cancellation line B is set to be equal to or shorter than the transport route length L**1** from the optical recognition sensor **18** to the inlet/outlet **12**. The reading target **103** is a portion including information related to a feature quantity that enables judgement of the type of the banknote **100**. The optical recognition sensor **18** is arranged at a position where the information of the total length LB of the reading target **103** in the transport direction can be read before the leading end **100***a* of the banknote **100** reaches the inlet/outlet **12**.
(31) The control unit **200** judges whether the banknote **100** is a banknote to be discharged on the basis of the feature quantity of a portion (the reading target **103**) of the banknote **100** having passed the optical recognition sensor **18** before the leading end **100***a* of the banknote **100** transported in the discharge direction protrudes to the outside of the banknote transport device **1** from the inlet/outlet **12**.
(32) The banknote **100** being discharged is a banknote stored in the banknote storage and has been authenticated at the time of storage into the banknote storage. That is, a banknote fed from the banknote storage is guaranteed to be a genuine banknote. Therefore, authentication of a banknote can be omitted when the banknote is discharged from the banknote storage. Information of the entire banknote is required to authenticate a banknote. On the other hand, the type (denomination) of a banknote can be judged with information of a portion of the banknote. About a half to a third of the total length LA of the banknote **100** in the transport direction is enough for the length LB of the reading target **103** in the transport direction.
(33) Accordingly, at the time of discharging the banknote **100** fed from the banknote storage, the banknote transport device **1** according to the present embodiment judges only the type of the banknote on the basis of information read from a portion (the reading target **103**) of the banknote **100** on the leading end side in the transport direction, and discharges the banknote **100** from the inlet/outlet **12** when the banknote is judged as a type of a banknote to be discharged.
(34) As illustrated in FIG. **3**(*c*), it is desirable that the range (the transport direction length LB) of the reading target **103** is set in such a manner that the leading end **100***a* of the banknote **100** does not go beyond the discharge cancellation line B also with a length (a portion indicated by C in FIG. **3**(*c*)) of the banknote **100** transported after the optical recognition sensor **18** reads the reading target **103** (after a back end **103***b* of the reading target **103** passes the optical recognition sensor **18**) and before the transport motor **35** is stopped or reversely rotated. With such setting, there is no need to stop the transport motor **35** when the discharge processing is not cancelled, and the discharge processing can be efficiently performed.
(35) FIG. **4**(*a*) is an example in which a nipping part **16***a* of the inlet/outlet roller pair **16** arranged nearby the inlet/outlet **12** is the discharge cancellation line B. In this example, the transport direction length LB of the reading target **103** is set to be equal to or shorter than a transport route length L**2** from the optical recognition sensor **18** to the nipping part **16***a*.
(36) The control unit **200** judges whether the banknote **100** is a banknote to be discharged on the basis of the feature quantity of a portion (the reading target **103**) of the banknote **100** having passed the optical recognition sensor **18** before the leading end **100***a* of the banknote **100** transported in the discharge direction reaches the nipping part **16***a*.
(37) Each of rollers constituting the inlet/outlet roller pair **16** protrudes into the transport route **10** to press against the other roller of the pair. Therefore, it is difficult to pinch the banknote **100**

located on the inner side of the banknote transport device **1** than the nipping part **16***a* with fingers or the like and take out the banknote **100**. That is, the nipping part **16***a* of the inlet/outlet roller pair **16** can function as extraction resisting means that provides resistance force against extraction of the banknote **100** from the inlet/outlet **12**, or extraction blocking means that blocks extraction of the banknote.

(38) The banknote transport device **1** illustrated in FIG. **4**(*b*) includes a shutter unit **60** that opens and closes the transport route **10**, on the transport route **10**. The shutter unit **60** is arranged at a position on the inner side of the banknote transport device **1** than the inlet/outlet tracking sensor **14** and nearby the inlet/outlet tracking sensor **14**. The shutter unit **60** moves between a position (a position indicated by a solid line in FIG. **4**(*b*)) to open the transport route **10** toward the inlet/outlet **12** and a position (a position indicated by a broken line in FIG. **4**(*b*)) to close the transport route **10**. When at the closed position, the shutter unit **60** functions an extraction blocking means that blocks extraction of the banknote **100** from the inlet/outlet **12**.

(39) As illustrated in FIG. **2**, for example, a solenoid **61** is connected to an output terminal of the control unit **200** as driving means that drives the shutter unit **60** to be opened and closed. The control unit **200** controls the solenoid **61** to move the shutter unit **60** to the opened position or the closed position according to a transported state, a transported position, or the like of the banknote **100**.

(40) FIG. **4**(*b*) is an example in which the shutter unit **60** is the discharge cancellation line B. In this example, the transport direction length LB of the reading target **103** is set to be equal to or shorter than a transport route length L**3** from the optical recognition sensor **18** to the shutter unit **60**.

(41) The control unit **200** judges whether the banknote **100** is a banknote to be discharged on the basis of the feature quantity of a portion (the reading target **103**) of the banknote **100** having passed the optical recognition sensor **18** before the leading end **100***a* of the banknote **100** transported in the discharge direction reaches the shutter unit **60**.

(42) The shutter unit **60** may be a switching device (a switching gate) that sets the transport destination of the banknote **100** to the inlet/outlet **12** or sets the transport destination of the banknote **100** to a branch route branching from the transport route **10**. In this case, when the switching device sets the transport destination of the banknote **100** to the branch route, the transport route **10** toward the inlet/outlet **12** is closed.

(43) <Depositing Flow>

(44) An operation in a case in which the banknote transport device **1** functions as a deposit device that introduces the banknote **100** into the device is explained with reference to a flowchart. FIG. **5** is a flowchart illustrating an operation at the time of banknote deposit.

(45) At Step S**101**, the control unit **200** waits to detect whether a banknote is input to the inlet/outlet **12**. When a banknote is input to the inlet/outlet **12** provided at one end of the transport route **10**, the inlet/outlet tracking sensor **14** detects insertion of the banknote **100** and sends a detection signal to the control unit **200**.

(46) At Step S**103**, the control unit **200** drives the transport motor **35** to transport the banknote along the transport route **10**.

(47) At Step S**105**, the control unit **200** turns on the optical recognition sensor **18**. The optical recognition sensor **18** reads an optical pattern (optical features) related to the transport direction total length LA of the banknote **100** being transported. The optical recognition sensor **18** sends the read recognition information to the control unit **200**.

(48) The control unit **200** as the authentication unit receives an output signal of the optical recognition sensor **18**, and authenticates the banknote as to whether the banknote being transported is a genuine banknote on the basis of feature quantity data (Step **107**). The control unit **200** as the paper sheet type judging unit judges the denomination of the banknote being transported on the basis of the feature quantity data.

(49) When judging that the banknote being transported is a genuine banknote (YES at Step S**107**),

the control unit **200** judges whether the communication port tracking sensor **30** has detected passage of a banknote at Step S**109**.
(50) When the communication port tracking sensor **30** has detected passage of a banknote (YES at Step S**109**), the control unit **200** stops the transport motor **35** at Step S**111**.
(51) When the banknote is stored in the banknote storage at Step S**113**, a series of processes ends.
(52) On the other hand, at Step S**107**, when the control unit **200** does not judge that the banknote being transported is a genuine banknote (NO at Step S**107**), the control unit **200** stops the transport motor **53** and reversely rotates the transport motor **53** (Steps S**115** and S**117**) to return the banknote to the inlet/outlet **12**.
(53) When the inlet/outlet tracking sensor **14** has detected passage of a banknote at Step S**119** (YES at Step **119**), the control unit **200** stops driving of the transport motor **35** (Step S**121**). When discharge of the banknote is complete (Step **123**), a series of processes ends.
(54) <Dispensing Flow>
(55) An operation in a case in which the banknote transport device **1** discharges a banknote is explained with reference to a flowchart. FIG. **6** is a flowchart illustrating an operation at the time of banknote discharge.
(56) At Step S**131**, the control unit **200** waits to detect whether a banknote is transported from the banknote storage. When a banknote is inserted to the communication port **32** provided at the other end of the transport route **10**, the communication port tracking sensor **30** detects insertion of the banknote **100** and sends a detection signal to the control unit **200**.
(57) At Step **133**, the control unit **200** drives the transport motor **35** to transport the banknote along the transport route **10**.
(58) At Step S**135**, the control unit **200** turns on the optical recognition sensor **18**. The optical recognition sensor **18** reads an optical pattern (optical features) associated with the reading target **103** of the banknote **100** being transported. The optical recognition sensor **18** sends the read recognition information to the control unit **200**.
(59) The control unit **200** acquires information related to the type of the banknote discharged from the banknote storage, from control means that controls the banknote storage, or the like. The control unit **200** receives an output signal from the optical recognition sensor **18**, and judges whether the banknote being transported is a type of a banknote to be discharged on the basis of the feature quantity data, that is, whether the feature quantity data matches the information related to the type of the banknote acquired from the control means that controls the banknote storage, or the like (Step **137**).
(60) Banknotes stored in the banknote storage have been already authenticated in accordance with the flow at the time of banknote introduction (FIG. **5**). The control unit **200** does not authenticate the banknote being transported on the premise that banknotes transported at the time of banknote discharge are genuine banknotes.
(61) When the control unit **200** judges that the banknote being transported is a type of a banknote to be discharged (YES at Step S**137**), the control unit **200** judges whether the inlet/outlet tracking sensor **14** has detected passage of a banknote at Step S**139**.
(62) When the inlet/outlet tracking sensor **14** has detected passage of the banknote (YES at Step S**139**), the control unit **200** stops the transport motor **35** at Step S**141**.
(63) When the banknote is discharged from the banknote transport device **1** at Step S**143**, a series of processes ends.
(64) On the other hand, at Step S**137**, when the control unit **200** does not judge that the banknote being transported is a type of a banknote to be discharged (NO at Step S**137**), the control unit **200** stops the transport motor **53** (Step S**145**). The banknote **100** is stopped before reaching the discharge cancellation line B.
(65) At Step S**147**, the control unit **200** reversely rotates the transport motor **35** to transport the banknote to the banknote storage.

(66) When the communication port tracking sensor **30** has detected passage of a banknote at Step S**149** (YES at Step S**149**), the control unit **200** stops driving the transport motor **35** (Step S**151**).
(67) When restorage of the banknote is complete (Step **153**), a series of processes ends.
(68) <Modification>
(69) While the banknote transport device **1** in the present embodiment is a device that performs introduction of a banknote into the device and discharge of a banknote to the outside of the device, the banknote transport device **1** may be a device that performs only discharge of a banknote to the outside of the device. In this case, it is assumed that banknotes stored in the banknote storage have been authenticated by some method at the time of storage into the banknote storage and that a banknote being discharged is guaranteed to be a genuine banknote from this point of view.
(70) While the banknote transport device **1** transports the banknote **100** along the long edge direction in the present example, the banknote **100** may be transported along the short edge direction.
(71) The optical recognition sensor **18** is arranged at a position where the sensor can judge each of plural types of banknotes that are supposed to be handled by the banknote handling device **1** at the time of discharge of the banknote.
(72) <Effect>
(73) According to the present embodiment, at the time of discharge of a banknote, whether the banknote is a banknote to be discharged is judged based on the feature quantity of a portion of the banknote on the leading end side in the transport direction. Since the transport direction length of a reading target used in the judgement of the banknote type is shorter than the total length of the banknote in the transport direction, the optical recognition sensor can be arranged close to the inlet/outlet. Therefore, the banknote transport device can be downsized.
(74) The amount of the feature quantity data used to judge the banknote type is smaller than that of the feature quantity data acquired from the entire banknote. Accordingly, the judging processing can be speeded up.
SUMMARY OF ACTIONS AND EFFECTS OF ASPECTS OF PRESENT INVENTION
First Aspect
(75) The present aspect is a paper sheet transport device (a banknote transport device **1**) including a paper sheet discharge port (an inlet/outlet **12**) for discharging a paper sheet (a banknote **100**) to outside the device, a recognition sensor (an optical recognition sensor **18**) that is arranged on a transport route **10** for the paper sheet toward the paper sheet discharge port to read a feature quantity of the paper sheet transported on the transport route, and a paper sheet type judging unit (a control unit **200**) that judges a type of the paper sheet on the basis of the read feature quantity of the paper sheet.
(76) In the paper sheet transport device, a transport route length L**1** from the paper sheet discharge port to the recognition sensor is configured to be shorter than a total length LA of the paper sheet in a transport direction, and the paper sheet type judging unit judges, at a time of discharging the paper sheet already authenticated, a type of the paper sheet on the basis of a feature quantity of a paper sheet portion (a reading target **103**) passing the recognition sensor before a leading end **100***a* of the paper sheet transported in a discharge direction protrudes to outside the device from the paper sheet discharge port. The paper sheet is transported from, for example, a paper sheet storage that stores paper sheets already authenticated.
(77) According to the present aspect, even when only the feature quantity of a portion of a paper sheet is acquired at a time of discharging the paper sheet, whether to discharge the paper sheet can be determined. That is, the type of a paper sheet can be judged when the feature quantity of a portion of the paper sheet, in this aspect, particularly the feature quantity of a portion of the paper sheet on the leading end side in the transport direction is acquired. Therefore, whether a paper sheet is a type of a paper sheet to be discharged can be judged with the feature quantity of a portion of the paper sheet.

(78) In the present aspect, the transport direction length of a paper sheet portion (a reading target) used to judge the type of the paper sheet is shorter than the total length of the paper sheet in the transport direction. Accordingly, the recognition sensor can be arranged close to the paper sheet discharge port. Therefore, the paper sheet transport device can be downsized.
(79) The amount of the feature quantity data used to judge the type of a paper sheet is smaller than that of the feature quantity data acquired from the entire paper sheet. Accordingly, the judging processing of the paper sheet type can be speeded up.
(80) As for a paper sheet being discharged from the paper sheet transport device, when the paper sheet is authenticated at the time of receiving the paper sheet into a paper sheet storage arranged in the back of the paper sheet transport device, the paper sheet is guaranteed to be a genuine paper sheet without authentication of the paper sheet being discharged. In this case, authentication of a paper sheet at the time of discharging the paper sheet can be omitted.
(81) The paper sheet transport device may include a function to transport a paper sheet introduced from outside the device to a paper sheet storage arranged in the back of the paper sheet transport device.
Second Aspect
(82) The present aspect is a paper sheet transport device (a banknote transport device **1**) including a paper sheet discharge port (an inlet/outlet **12**) for discharging a paper sheet (a banknote **100**) to outside the device, a recognition sensor (an optical recognition sensor **18**) that is arranged on a transport route **10** for the paper sheet toward the paper sheet discharge port to read a feature quantity of the paper sheet transported on the transport route, and a paper sheet type judging unit (a control unit **200**) that judges a type of the paper sheet on the basis of the read feature quantity of the paper sheet.
(83) The paper sheet transport device includes a discharge roller pair (an inlet/outlet roller pair **16**) for banknote transport, that is arranged nearby the paper sheet discharge port, a transport route length L**2** from the discharge roller pair to the recognition sensor is configured to be shorter than a total length LA of the paper sheet in a transport direction, and the paper sheet type judging unit judges, at a time of discharging the paper sheet already authenticated, a type of the paper sheet on the basis of a feature quantity of a paper sheet portion (a reading target **103**) passing the recognition sensor before a leading end **100***a* of the paper sheet transported in a discharge direction reaches a nipping part **16***a* of the discharge roller pair. The paper sheet is transported from, for example, a paper sheet storage that stores paper sheets already authenticated.
(84) According to the present aspect, identical effects as those of the first aspect can be attained.
Third Aspect
(85) The present aspect is a paper sheet transport device (a banknote transport device **1**) including a paper sheet discharge port (an inlet/outlet **12**) for discharging a paper sheet (a banknote **100**) to outside the device, a recognition sensor (an optical recognition sensor **18**) that is arranged on a transport route **10** for the paper sheet toward the paper sheet discharge port to read a feature quantity of the paper sheet transported on the transport route, and a paper sheet type judging unit (a control unit **200**) that judges a type of the paper sheet on the basis of the read feature quantity of the paper sheet.
(86) The paper sheet transport device includes a shutter unit **60** that is arranged nearby the paper sheet discharge port and moves between a position to open the transport route and a position to close the transport route, a transport route length L**3** from the shutter unit to the recognition sensor is configured to be shorter than a total length LA of the paper sheet in a transport direction, and the paper sheet type judging unit judges, at a time of discharging the paper sheet already authenticated, a type of the paper sheet on the basis of a feature quantity of a paper sheet portion (a reading target **103**) passing the recognition sensor before a leading end **100***a* of the paper sheet transported in a discharge direction reaches the shutter unit. The paper sheet is transported from, for example, a paper sheet storage that stores paper sheets already authenticated.

(87) According to the present aspect, identical effects as those of the first aspect can be attained.
Fourth Aspect
(88) In the paper sheet transport device (the banknote transport device **1**) according to the present aspect, the paper sheet discharge port (the inlet/outlet **12**) is a paper sheet inlet/outlet for discharging the paper sheet (the banknote **100**) to outside the device and receiving the paper sheet into the device, the device includes an authentication unit (a control unit **200**) that authenticates the paper sheet on the basis of the feature quantity of the paper sheet read by the recognition sensor (the optical recognition sensor **18**), and the authentication unit authenticates the paper sheet at the time of receiving the paper sheet into the device from the paper sheet inlet/outlet, on the basis of the feature quantity of the paper sheet read by the recognition sensor.
(89) At the time of receiving a paper sheet from the paper sheet inlet/outlet, the paper sheet transport device authenticates the paper sheet. The paper sheet transport device transports a paper sheet introduced from the paper sheet inlet/outlet to a paper sheet storage that is arranged in the back of the paper sheet transport device. Therefore, at the time when a paper sheet transported from the paper sheet storage is discharged from the paper sheet inlet/outlet, authenticity of the paper sheet is guaranteed even when authentication of a paper sheet is omitted.
Fifth Aspect
(90) The present aspect is a paper sheet handling device including the paper sheet transport device (the banknote transport device **1**) according to any one of the first to fourth aspects.
(91) According to the present aspect, identical effects as those of the first to fourth aspects can be attained.
REFERENCE SIGNS LIST
(92) L**1** to L**3** transport route length, LA total length in transport direction of banknote, LB transport direction length of reading target of banknote, **1** banknote transport device, **3** lower unit, **4** upper unit, **10** transport route, **12** inlet/outlet, **14** inlet/outlet tracking sensor, **16** inlet/outlet roller pair, **16***a* nipping part, **18** optical recognition sensor, **18***a* light-emitting element, **20** relay roller pair, **22** tracking sensor, **24** fraud prevention mechanism, **26** tracking sensor, **28** communication port roller pair, **30** communication port tracking sensor, **32** communication port, **35** transport motor, **50** opening/closing member, **53** transport motor, **60** shutter unit, **61** solenoid, **100** banknote, **100***a* leading end, **101** leading end portion, **103** reading target, **103***b* back end, **120** fraud preventing motor, **200** control unit.

Claims

1. A paper sheet transport device comprising a paper sheet discharge port for discharging a paper sheet to outside the device, a recognition sensor that is arranged on a transport route for the paper sheet toward the paper sheet discharge port to read a feature quantity of the paper sheet transported on the transport route, and a paper sheet type judging unit that judges a type of the paper sheet on a basis of the read feature quantity of the paper sheet, wherein a transport route length from the paper sheet discharge port to the recognition sensor is configured to be shorter than a total length of the paper sheet in a transport direction, and the paper sheet type judging unit judges, at a time of discharging the paper sheet already authenticated, a type of the paper sheet on a basis of a feature quantity of a paper sheet portion passing the recognition sensor before a leading end of the paper sheet transported in a discharge direction protrudes to outside the device from the paper sheet discharge port.
2. The paper sheet transport device according to claim 1, wherein the paper sheet discharge port is a paper sheet inlet/outlet for discharging the paper sheet to outside the device and receiving the paper sheet into the device, the device includes an authentication unit that authenticates the paper sheet on a basis of the feature quantity of the paper sheet read by the recognition sensor, and the authentication unit authenticates the paper sheet at a time of receiving the paper sheet into the

device from the paper sheet inlet/outlet, on a basis of the feature quantity of the paper sheet read by the recognition sensor.

3. A paper sheet handling device comprising the paper sheet transport device according to claim 2.

4. A paper sheet handling device comprising the paper sheet transport device according to claim 1.

5. A paper sheet transport device comprising a paper sheet discharge port for discharging a paper sheet to outside the device, a recognition sensor that is arranged on a transport route for the paper sheet toward the paper sheet discharge port to read a feature quantity of the paper sheet transported on the transport route, and a paper sheet type judging unit that judges a type of the paper sheet on a basis of the read feature quantity of the paper sheet, the device comprising a discharge roller pair for banknote transport, that is arranged nearby the paper sheet discharge port, wherein a transport route length from the discharge roller pair to the recognition sensor is configured to be shorter than a total length of the paper sheet in a transport direction, and the paper sheet type judging unit judges, at a time of discharging the paper sheet already authenticated, a type of the paper sheet on a basis of a feature quantity of a paper sheet portion passing the recognition sensor before a leading end of the paper sheet transported in a discharge direction reaches a nipping part of the discharge roller pair.

6. The paper sheet transport device according to claim 5, wherein the paper sheet discharge port is a paper sheet inlet/outlet for discharging the paper sheet to outside the device and receiving the paper sheet into the device, the device includes an authentication unit that authenticates the paper sheet on a basis of the feature quantity of the paper sheet read by the recognition sensor, and the authentication unit authenticates the paper sheet at a time of receiving the paper sheet into the device from the paper sheet inlet/outlet, on a basis of the feature quantity of the paper sheet read by the recognition sensor.

7. A paper sheet handling device comprising the paper sheet transport device according to claim 6.

8. A paper sheet handling device comprising the paper sheet transport device according to claim 5.

9. A paper sheet transport device comprising a paper sheet discharge port for discharging a paper sheet to outside the device, a recognition sensor that is arranged on a transport route for the paper sheet toward the paper sheet discharge port to read a feature quantity of the paper sheet transported on the transport route, and a paper sheet type judging unit that judges a type of the paper sheet on a basis of the read feature quantity of the paper sheet, the device comprising a shutter unit that is arranged nearby the paper sheet discharge port and moves between a position to open the transport route and a position to close the transport route, wherein a transport route length from the shutter unit to the recognition sensor is configured to be shorter than a total length of the paper sheet in a transport direction, and the paper sheet type judging unit judges, at a time of discharging the paper sheet already authenticated, a type of the paper sheet on a basis of a feature quantity of a paper sheet portion passing the recognition sensor before a leading end of the paper sheet transported in a discharge direction reaches the shutter unit.

10. The paper sheet transport device according to claim 9, wherein the paper sheet discharge port is a paper sheet inlet/outlet for discharging the paper sheet to outside the device and receiving the paper sheet into the device, the device includes an authentication unit that authenticates the paper sheet on a basis of the feature quantity of the paper sheet read by the recognition sensor, and the authentication unit authenticates the paper sheet at a time of receiving the paper sheet into the device from the paper sheet inlet/outlet, on a basis of the feature quantity of the paper sheet read by the recognition sensor.

11. A paper sheet handling device comprising the paper sheet transport device according to claim 10.

12. A paper sheet handling device comprising the paper sheet transport device according to claim 9.